ActInf Livestream #026.2 ~ “Bayesian Mechanics for Stationary Processes”

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hello everyone welcome to the active inference lab we're here on august 17 2021 in live stream number 26.2 on the second discussion that we're having as a group on the paper bayesian mechanics for stationary processes welcome to the active inference lab everyone we are a participatory online lab that is communicating learning and practicing applied active inference you can find us at the links here on this page this is recorded in an archived live stream so please provide us with feedback so that we can improve on our work all backgrounds and perspectives are welcome here and we'll be following video etiquette for live streams so we can use the raise hand feature in jitsi this short link has the upcoming live stream discussions here we are on august 17th in our last discussion on this paper 26.2 next week we'll be heading into a new paper and we still have some openings in the coming weeks so if anyone has a suggestion or an author to contact or wants to discuss their own paper just let us know and we can try to make that happen today in acting stream number 26.2 the goal is to continue learning and discussing and unpacking this awesome paper that we've been thinking about for the last few weeks bayesian mechanics for stationary processes and we're really appreciative that the first author is here to join with us and co-learn with us have fun with us and this video is just going to be wherever we want to take it so we have some questions prepared but especially if anyone in the live chat has questions that'd be awesome and we can definitely address them so we can just start with a quick introduction round and feel free just to say hello or to answer one of these questions like what's something that you liked or remembered about the paper and what's something that you're wondering about or want to have resolved by the end of today so i'm daniel i'm a postdoctoral researcher in california and something that over the last week made me excited to have this discussion was i was speaking with a physicist friend someone who's working with lasers and oscilloscopes and stuff and i brought up this paper and they it turned out that they actually were using pid control in their research on a day-to-day basis but actually it was new for them to hear about how it was connected to some of these broader topics but as soon as i brought out the physical printed paper then they were just so excited and it was cool to see how they had all this practical experience working with pid and then they instantly just saw that there was kind of this crossover point into some broader questions that also they were curious about but the day-to-day practice actually hadn't brought them to those general questions so it was a and i learned a lot about pid control and about actually how it plays out with the knobs and everything like that i'll pass to dean yeah good morning um yeah i i came away from the paper again trying to figure out how to create some sort of a coherent narrative around the the physical math and the statistical math that part for me really got me going away and thinking a little bit because i think they they go in different directions but i do think that they can work in concerts so yeah i'd be really interested to see how that might play out in terms of what what our our guest here um thinks the directions this might be able to go i'll pass it to stephen hello i'm stephen i'm based in toronto um i work a lot with community development and uh sort of spatial meaning making so inferences of different kinds i'm curious how some of the ways that the mathematics have been done on this paper can help broaden some of the questions around ergodicity and non-ergodicity in uh that's a bit of a thorn sometimes a weapon in the inactivist active influence dialogue so um yeah any and that's something i'll be interested in and um i'm going to pass this over to dave so i think dave doesn't have audio but dave thanks for joining and any questions in the youtube live chat is awesome so thanks to everyone who's joined and passed to lance thanks daniel and thanks for inviting me again uh sam nance i'm a phd student at imperial and ucl and i'm really excited to be here and to engage in this discussion i'm really looking forward to feedback about the paper seeing what's interesting to to pursue in future work um so yeah really looking forward to to share this paper with you and and talk about it so maybe one lead-in question before we jump to the content what stage in your phd are you and is this like the meat and potatoes chapter one or is this um a little cul-de-sac that you found yourself having to explore in the course of another research project um yeah great question so i'm right in the middle of my phd i'm two years in to still have two years to go and um ideally i would like to pursue this work but so i have two supervisors one is a mathematician another one is a neuroscientist and so really the the goal is to is to have some kind of mathematical theory of how adaptive systems work so i think this is really the the first chapter i want to push that further but then there are also other projects that take up my time so i'm trying to make all that work together awesome so for this dot two we have some questions prepared some topics that we know we to talk about but anyone who asks a question in a live chat will definitely have time to address that the paper that we've been talking about is this bayesian mechanics for stationary processes paper and in the dot zero and the dot one there's more information on the primary aims and claims so check those out if you haven't already we're just going to take one more look at the road map where we've been where we're going remember we were talking about markov blankets and basic mechanics how does active inference have similarities and differences from other ways to model dynamical systems and let's just go straight to the first question that we have which is going to help us review where we've been and then also as a jumping off point for some of these extended questions that we've already written so a colleague asked me and said i think it would be good to expand on the differences of the paradigms in section 3. and those are section b c and d related to a posteriori estimation after the fact predictive processing and variational base so perhaps lance first and then anyone else who wants to give a take or ask a follow-up question what are the similarities and the differences here and why were they structured in this order is this the chronological order is this most specific to least specific or what is the ordering and the meaning here right um yeah so so the it goes from the the simplest to the most sophisticated um so uh adversary estimation is a special case of predictive processing which is specific special case of variational base and so the idea is if you if you receive some piece of data so for example if you see an image um with your eyes you're gonna try to infer what's going on outside in the environment and so if you do a posterior estimation what you're trying to guess is what is the most likely thing um like what what is the most likely thing that's going on in the environment so it could be if you if you see an image of a chair you you would say oh it's a chair but you wouldn't encode any any kind of uncertainty oh it might be like an armchair for example or something a bit different so in a posterior estimation you try to guess what's going on outside but there's no uncertainty then predictive processing you encode the uncertainty so mathematically that's called the variance or the ov precision and then variation of bayes it's uh it's more sophisticated so you actually encode uh an arbitrary distribution uh on external states um so it could be oh it's very likely that it's an armchair and it might also be very likely that it's a cat so that this example doesn't make sense but you you could imagine that there's two things that are very different that could be equally likely so for example if you see an illusion it could be something but also something very different so you would imagine these two things of having a high probability of being there and then other things are having low probability so this is really um the the most general thing uh and also as a side comment so rational base um like a lot of machine learning algorithms you can see them as special cases of variational base so i think this is really the most general way to think about inference and and this is really at the heart of the free energy principle and i think this is why people have been describing the brain as performing variation of base because it's really the most general way to think about this kind of problem of inference awesome well before i give a comment on that what made you say that it was at the heart of right so the free energy principle says that the brain is doing a variational base and and then all this work i mean this paper in particular but like the last decade people have been trying to justify why would the free energy principle apply actually um and so and so this is part of uh this paper is part of a line of work where we try to examine physical systems or like adaptive systems and see whether we can describe them as performing variational base great so first stephen with a question and then anyone else with a question thanks that's really helpful that explanation and i was wondering is it a case then that you can go only so far with say predictive processing um and then to get to variational bays you could and to bring in that uncertainty you kind of need a free energy principle solution to be able to bring in these more unknown models and to go beyond variational bays at some point that's when you'd have to bring in the active inference to it would that be a kind of the way you would think about it yeah yeah we i would think about it like this um i would say uh like loosely speaking the term predictive processing i think is also meant uh to refer to variational base but at least a variational base that's like a proper mathematical definition of what it is and then predictive processing it's more like oh we're predicting something so it's not that clear how how i mean there's no clear definition of predictive processing uh but in this paper predictive processing is meant as predictive coding as in like the papers by ryo and ballard where they looked at the visual system and then they looked at neurons in the cortex as forming predictions and suppressing prediction errors um so yeah um if you if you only do predictive processing in in this sense i i think you're you're limited and variational base and active inference is really like the the next generation because you can accommodate all sorts of models and um and yeah i don't i don't think it i mean i i cannot see at least uh how you would go beyond that thank you so dean with a raised hand and then anyone else no that's i was curious so it seems like the math is able to help us understand the difference between something that's really highly defined so that kind of a posterior posteriori and then the sort of the fuzzy where the very old variational base part sort of gives us something a little more a little less defined in terms of the edges and so i was just curious does the brain kind of go back and forth to to some degree mathematically speaking between that which is really clear and really has a hard edge mathematically and and those things which statistically seem to distribute within the within the parameter set yeah i mean i think all these three um like a posterior estimation as well as variational base there's uh displausible explanations for how the brain would implement them okay and so so the idea the idea for um variational base or predictive processing is that so the brain would be encoding some kind of uncertainty and so in the literature people say that the the mean firing rate of a neural population would encode the posterior estimates so what you think is actually out there and then the variance of the firing rates would encode your uncertainty so i think so i think there's possible plausible theories for how the brain would implement all of these things and then i think the that's the nice thing at least with active inference is that we try to develop models that are consistent with our understanding of the brain and as our understanding of the brain evolves we kind of like refine these models um so so i think we're definitely working towards uh models that that do explain at least i mean so far it's very coarse grain and very simplistic but we're working towards that could be implemented in the brain so i think like these um very very abstract very general form of variational base could be implemented in the brain one quick follow-up is you mentioned how the mean firing rate corresponds to this first a posteriority estimation which is a mean estimation and then there's the variance in the firing rate so what measurements or what does it look like to implement variational bays in the brain like what measurements are we making what kinds of analyses would we make on those just like mean and variance well so so the idea i mean the very simple idea is that you would get some sensory stimulus so you you get some some impression on your retina uh some neurons in your retina with fire uh into the into the visual cortex and then um the the the premise is that your brain has embodies them a generative model of its environment so so it's able it has learned how to recognize uh how to how to infer things in the environment from its sensations and so the the stimulus that would be that would go into into v1 and all these other parts of visual cortex and then the neurons in in the visual cortex would uh align their means firing rates with respect to oh it's an armchair or it's something different um so maybe there's a neural population that encores armchair and there's a neural population that encodes something there's also a lot of studies for specific neurons like one neuron encoding one particular stimulus so these um there's these guys at mit that show that there's actually a bill clinton neuron uh that fires whenever you see bill clinton so you can be very and um and yeah but if it's uncertain that it's bill clinton your your bill clinton neuron would like fire yeah i guess would have a lot of variance in its firing rate okay awesome so welcome blue with a raised hand go for it and then um we'll continue and also steven so go ahead blue um i am blue knight and i'm an independent research consultant from new mexico and i have a question on your paper and it's kind of something that i've been questioning a lot lately with regards to the free energy principle and the requirement for ergodicity um and my question is um in your paper you said that the um you know this the outline even in outline it says you know you you partition the world with the markov blanket and then you equip the partition with stochastic dynamics and so i'm wondering how these stochastic dynamics relate to organicity if at all and if that fulfills the requirement for um ergonomy as necessary by the fbp right yeah that that's a great question um yeah i think i think stephen also also wanted to talk about that so it's great that you bring this up yeah so so the idea of these stochastic dynamics is that we don't assume that dynamics are deterministic and it makes sense because um you can describe planets as having deterministic dynamics but when you talk about biological systems things are a lot more messy and if you're not modeling them at the molecular level which you which you never are things are stochastic so then the the assumption on our system is is that it's stationary and this is really what it means to to be adaptive so you're preserving some kind of steady state but we never assume ergodicity so there's actually um a deep um so stationary and ergodicity are not the same so we assume stationarity but we don't assume ergodicity and so if daniel you could go to the appendix b there's um there's a figure i think which would be the best way to to illustrate a bit below yeah perfect here in this appendix we we illustrate uh some dynamics that are ergodic and some dynamics that are not ergotic if you if you look at the uh top left top and bottom left panels these dynamics are ergotic and and so i guess the intuitive definition of ergodicity is that here you have the the steady state in blue so so regions that are very blue are regions of high probability mass under the state state of regions where you your dynamic is likely to be in and regions are less blue you're less likely to be in now ergodicity means that if you start your dynamic anywhere and you and you let it run for a sufficient amount of time so like you observe your biological system for a very long time um your dynamic would basically sample the whole distribution it would go i mean if you if you give it enough time it would go really far away and then it will come back and it will do all sorts of things as you can see in the bottom left and bottom right um so this is ergotic now in the in the in the bottom oh no in the top left and and bottom left it's it's ergotic and then the bottom right it's not ergotic and so um so as you can see if you start the dynamic on the bottom right anywhere it will go around an orbit they won't sample the whole distribution so it's still confined to like a a small region and and so this this uh so the the paper still applies for these kind of dynamics see i don't i don't know does that help or do you or was that not very clear it's so such a follow-up the paper does apply in situations where it's not that's what you said right right we yeah we don't need aerodynacity well so even within like a steady state um like so biological systems you know operate within homeostasis but it's a range right so so it's not like a constant it's not like you know purely we're not at 37 degrees all the time right like so there's some fluctuation all the time within that steady state and and so i just want to kind of how you rectify those concepts like is it like a restricting of the the state space or or how does that work so so the this idea of homeostasis that's that's really the the steady state assumption so for example um in the state space maybe you would have one one uh one coordinate that describes the temperature of your of your biological system and if it's a human um then you will have regions of very high probability so here regions in blue around 37 degrees and then regions that are less blue uh like a bit further away uh and so the dynamic i mean your your system can evolve in in all of these regions just like on average you will be more often around 37 degrees just because that's how we are and then but sometimes it will be higher and sometimes it will be lower and and so this is all allowed um and so i guess the this ergodicity assumption so then if you assumed that the biological system was ergodic then this would imply that if you if you leave the biological system like run for a sufficiently long time like you observe it for a very long time it will basically um it will sample your whole distribution of temperatures so so maybe so so maybe like the the temperature that um a human body is allowed to have are between like 36 and 39 let's just say for for simplicity and so ergodicity would say um yeah if i if i observed my biological system for a sufficiently long time i would see my biological system at all these temperatures and i think maybe maybe that would happen so maybe our uh maybe temperature is like uh we're ergotic with respect to temperature but maybe there's other things in which we aren't ergotic and i think the the previous papers on the free energy principle they for simplicity they always took like um they always assumed some kind of dynamic and it turned out to be ergotic and so in here we we're not making that assumption i'm sorry i'm going to just follow with one more question daniel is that okay yeah um you know it's great right if you're talking about one human system right like my mark on blanket the partitions like internal and external states and so i have my one human system or even multiple human systems like we will operate in the same kind of steady state based range but but if we're talking really about um you know collections and how kind of collective dynamics and overlapping markup blankets do you think it's mathematically possible to extend the state space like i can operate you know from 36 to 39 that's an okay temperature for me but like a reptile has a very different kind of temperature range but clearly like we have some interaction i have interaction with the reptile and so there's some some emergent dynamic between me and some reptile future right right like so can we define a state space like would it be mathematically like acceptable or is it like offensive mathematically like impossible to define a state space where i'm allowed to sample all of these temperature ranges with my markup blanket but some other feature in some kind of collective dynamics is not like they have a different range like is that still organicity because it's not clearly i'm not sampling the entire state space i'm sampling my state space but like is that confounding in some way i mean it is in my brain that's why i'm asking yeah that that's a really good question um so so this is all fine so so we can do everything you just said and so when we um so actually when when we say oh we so we we always sample a distribution we're not i mean and then the distribution is on the safe space so we're sampling the state space but like if your distribution is confined to a region of state space so let's say as a human um i cannot possibly have any bodily temperatures below 36 degrees then your then you would sample only things are above 36 degrees um so i think you would have your own uh steady state your own stationary distribution with temperatures around 37 degrees and then if there's a if there's a snake also in the in this environment that you're modeling then we'll have a another very different steady state i think it really uh your question really speaks to the possible extensions of these of this because in this paper we just say oh there's like an internal state space which would be what's inside and then there's an external state space which would be what's outside but then um really what what people would need to do now is to say oh but my actually there's a lot of very interesting things in my environment there may be snakes there may be other kinds of animals and they all have their own markov blanket and they all have their own steady state so then what you can do is to partition your external state space into all of these things that might be around you and then instead of having some kind of um performing some kind of simple inference as we have in this paper you would have some more structure inference a more structured inference because part of your inference would be about what's just in front of you part of your inference would be about your dog that's right over there and you would have a lot of by carving your external state space into different things with their own steady state you would actually have a an inference that's much more structured because you would infer all these things all these different things that are in your environment so this can all be accommodated in this paper but then you would definitely need another paper to like make that explicit and say some really interesting things because yeah this is really um the way to go but the the mathematics doesn't need doesn't need extension it's just like um yeah it would just be like partition the state space and then yeah we can have all that that you spoke about thank you great and also one piece so then stephen's question is it's it's not as if something being ergodic or not is equivalent to it being easy or simple or tractable like looking at these distributions um or trajectories here the one that you characterized as non-ergotic not sampling um basically across the state space the bottom right this one is the one that in the caption it says purely conservative dynamics lower right panel are reminiscent of the trajectories of massive bodies whose random fluctuations are negligible so it's actually not like it's um maybe this will speak to steven's questions about what is a thorn and and what's a rose because this is a non-ergotic system that's massively predictable you can do it with a you know a gear maybe that relates to some of these criticisms about well we're only working with a limited set of ergotic processes it's like okay but that's the squiggly one that's dancing everywhere the stochastic process that we care about whereas including non-ergotic processes some of these non-organic processes might just be going around in an oval so it won't be a challenge for the math at all something that works on the stochastic squiggle is going to have actually an easier time on this oval one so stephen and then anyone else with a question yeah thanks um yeah i suppose actually tying into what you just said there daniel the that there's this question with um how much we need to be able to sample over big state spaces so it might well be that i don't know the chemicals of a cell there with molecules vibrating millions of times a second and there may be a you know a quick ongoing way to sort of get a feel for the ergodic state space now if you watch a cricket ball come towards you you know you don't get that chance to do that so but you might want to sort of maintain a trajec some sort of trajectory um that can still be um tractable i suppose so i think this is this is actually quite quite useful um i suppose the question then comes in how what's the sort of in-betweens you know where does um approximation science come in more and more to allow us to sort of work with not knowing um and work with something like um these solenoidal flows in a more fuzzy way um because i am assuming that if you've got less chances to sample your state space solenoidal might be more useful to get towards something tractable or maybe i'm wrong but i wonder what your thoughts are on that right um so actually solenoidal um i mean so in in this figure here uh on the bottom left you have like no solenoidal flow and then the bottom right there's just so the node or flow um so i would say all of these two like very special cases uh they don't look like biological systems at all and they're also the the simplest ones to to understand mathematically and to study mathematically and actually when it gets really interesting and when it looks a lot more like a biological system is on the top left and that's harder to to assess mathematically um and so as you can see it's quite i mean i think the dynamic on the top left is quite nice because on the one hand you have dissipation like you have some squiggly lines you have randomness on the other hand thanks to the solenoidal flow you have some kind of exploration and also some kind of like circular patterns so for example in um and this is actually how how you correct there is what's called a non-equilibrium steady state but without without going into the details um as humans as and as animals we have a lot of circular patterns they are not purely deterministic so for example we all have a circadian rhythm um and we all have all sorts of other um yeah circular patterns and i think yeah to account for that uh you really need to be to be on the top left um so i think the [Music] i mean one of the next steps to this research is right right now we just assumed that the dynamic is stationary and so all this applies to like a humongous range of dynamics and a lot of them don't look like biological systems at all and i think the next step would be to to restrict ourselves to dynamics that look a lot more like that like biological systems and then see what what else can we say and and i bet there's some dynamics that look a lot like very intelligent systems and some that also look like maybe lifelike systems but they are not so intelligent so the question is what kind of parameters what kind of dynamics do you need um to characterize something that explores its environment uh something that makes decisions and all sorts of things yeah and and that's going to be a challenge and also i mean just speaking about biological systems i think biological systems have a lot of solenoidal flow and they do have dissipation but they don't have so much dissipation so you would have something like the top left but with like huge you have huge like a lot of expiration like you would go a lot further than that in your state space and then you would have a bit of i mean you would have a lot of squiggly lines but it would be yeah there would be a lot more like exploration than there is dissipation yes stephen go forward yeah and and you have both can combine can't so gradient sin and solenoidal flow can kind of be playing at the same time i'm wondering if someone was to bring attention to a goal or a type of attention maybe more pragmatic attention or an epistemic attention or a risk adverse attention whether it might be that rather than trying to deal with the whole state space maybe that that general process carries on but in it tries to introduce a solenoidal you know um type of flow which in some ways you could imagine being more tractable to a biological system if it's like bioelectricity um there's you know the types of dynamics which may be are more tractable around a certain trajectory um than some um so you've the these the these dynamics could combine in different regimes of attention yeah definitely um and i think but i think to to properly account for attention we we need to go beyond these continuous state spaces and do what people um um and basically extend to these standard models of active inference where in in addition to have some continuous state space maybe like for movements and this kind of thing you also have a discrete state space and then you would have a dynamic evolving in your discrete state space and maybe like some state would be oh i'm attending over there and some other state would be unattending over there so i think attention is really something um that can be modeled with a discrete state space and then it would i don't think it would be so hard and i mean at least seems doable to extend all the math here to like models uh yeah that have some like continuous state space maybe for movement or or or other kinds of things like maybe like blood glucose concentration is also something that evolves on a continuous scale there's also a lot of a lot of things in our mind that are discrete so is this red yes or no is it raining yes or no and to account for all that we need uh we need a discrete state space as well awesome thank you lance welcome scott i see your hand raised so go for the question and then anyone else with a raised hand thank you i came in a little late um my apologies is fascinating the solenoidal flow when that gets set up is it possible that he had multiple state spaces and might because of the solenoidal flow might it have a periodicity in each of the state spaces so that you might be able to use fourier transforms to unpack multiple state spaces in a biological system so could you use having that periodicity does that allow for a fourier type analysis to start to unpack state spaces potentially when you have multiple state spaces and you're trying to distinguish them yeah that's a that's a really good point so i know um i know a paper actually that that analyzes the solenoidal flow in terms of the fourier transform um i haven't really really thought about it i think i mean if if you get rid of the dissipative part so if you get rid of the squiggly lines and you're just in the bottom right then you can definitely use fourier transforms but then also the dynamic on the bottom right is not is not representative of so then what you can do when you're on the top left which is really the interesting case is analyze the time irreversible part so the solenoid will flow with the fourier transform and then so you so you do this hem holes decomposition that we're talking about in the appendix uh you decompose your dynamic into yeah so it's the equation a2 yeah yeah perfect so you decompose your uh your drift b so so b is the direction in which your dynamic is moving um on average into reversible and irreversible parts the irreversible part is the solenoid overflow which you can analyze using fourier transform but then you need to like figure out how to accommodate the reversible part with it so i can share i can share the paper if you're interested but yeah it seems like like they're they're able to to like do do some things with that technique uh but it's also pretty pretty involved to be honest a question on the time reversible and irreversible what does that mean and in a world where it seems like we're moving forward through time only what exactly does that mean that there's something that's invariant under time reversible versus something that changes sign under time reversal like if we can't get the time machine or put the process in a time machine why does it matter that it's time reversible or how does that fall out of the equations well yeah so so all this is really to emphasize that we're not assuming the dynamic to be time reversible and and as you say everything in the world around us i mean most things are time irreversible so uh if you go back to the to the figure just below so the movement of planets is really something that's irreversible i mean the the earth turns around the sun in one direction and if you are to rewind time it will turn around the sun in the other direction so that so in this case you can really tell the difference between time as time goes forward and as time goes backward and that's really what's meant by time irreversible um but now if you look like a other cup of coffee and the coffee is still um the the molecules the coffee molecules or whatever molecules you have in that cup they're still bouncing with each other and that's that's something that's time reversible because if you look at the cup of coffee um if you if i give you a movie of the cup of coffee sitting still you can watch the movie uh as time goes forward or backward and uh and you wouldn't be able to tell the difference so that's something that's reversible and so as you can see from this example actually things that are reversible aren't interesting at all from at least a biological perspective and there's a there's so much literature um yeah about one of the uh i mean one of the fundamental things of being alive is being time irreversible it's about like eating and producing waste um it's about yeah consuming all that kind of thing all that kind of stuff um produces entropy and is time um but then time so time irreversibility it's something that's necessary to be alive but it's not it's not sufficient um so for example if i give you um a convection cell though that would be time irreversible but it won't it won't be alive so any any kind of cycle that's more or less periodic um so maybe if you if you see like a cyclone a cyclone is time irreversible because it's like spinning one way and not the other so if you run out of time it would be spinning the other way so it's different so it's time irreversible but that's not alive um and so there's many many things like that time irreversibility um yeah creation of entropy there's a few things like that that characterize biological systems and i think as we as we go on with this work we'll we'll try to like specialize to only those dynamics that satisfy all these requirements and eventually things like oh we need systems that are able to reproduce for example like a marker blanket that is able to divide into two markup blankets for example when a when a cell decomposes into into two cells so so this is really the beginning and and yeah in the next years i think this community is really going to be able to to like address all these challenges and like all these characterizations of living systems to to gradually evolve towards models that are yeah more realistic i would say i'll have more to say on it but scott go for it so a couple of things there's a book called life's ratchet which is kind of interesting just on that um the idea of that irreversibility and that value of that the idea is that life kind of ratchets in these forms that work with the externality i haven't revisited that since i've been exposed to active inference but it might be interesting to take another look at life's ratchet also one of the things i like that discussion of living forms and what's necessary uh prerequisites to living forms one of the working definitions i'll share that i'm using now and i think let me tell you why i'm using it is um that life is auto catalytic and entropy secreting now my kids claim the i just wanted to say secretion in my definition because it's gushy that was my my kids used to say but um auto catalytic and entropy secreting so notice i didn't say it had to be embodied in a specific form or specific matrix or context it could be a market it could be a markov you know a lot of different things so um so the reason i share that here it's interesting because active inference we have this definition of life like corollas linnaeus you look at it you see if it has feathers and you see if it has blood and stuff like that but now we're going towards mathematical definitions of living things and that's going to be very interesting because i think what when we have a working definition that works for biologic forms so again my working definition right now is auto catalytic um then i can start to say okay if that works for biologic forms let's say that did work and maybe there's problems with whatever but if it works then you can start saying okay what other systems happen to be auto catalytic and entropy secreting and therefore we shall now call living and maybe we can start to look at them and make observations about those systems and what i'm thinking about i'm a guy who's a lawyer and in finance for years so i'm looking at this and saying okay you look at these systems biological systems came up with these markets and these other systems that we have to do things in the world maybe these systems are describable with biological framings and so that's why the kind of work you're doing feels like it as it starts to reveal some more general definitions of living forms taking those definitions and then reapplying them back out into the world and saying okay are there living forms that have escaped our detection as living forms and basically are they is there scale independence are there fractal elements what are the things that we can now describe in those things that we previously had not thought of as biology but in fact are describable as biology that's very exciting to me so i just wanted to say that the kind of things you were describing there and that kind of working way back down into a model of life based on time irreversibility it's very interesting i think thanks and very interesting paper overall i'm looking forward to diving into this and the other that other paper you were mentioning also thank you cool yeah yeah thanks a lot that that's really great so to tie this from the some of the physical examples we've been talking about like the diffusion processes to the physics of information and information flows the big thing we've been talking about is this statistical physics and this bayesian mechanics and we heard a little bit earlier how we could talk about the moments like the first moment the simplest form the a posteriori estimation was the mean and then the second moment of a statistical distribution is the variance and there's also higher statistical moments that describe higher and higher aspects of an informational distribution there's also the concept in physics of moments like the position and the velocity in higher and higher physical moments and so the mapping is that potentially these physical moments that would describe components of a physical flow that could also be decomposed using the helmholtz decomposition might also apply to some of these informational flows and decompositions so we could use those analogies like the the coffee mixing and the steady state of the coffee and then think about how a statistical distribution is playing by similar rules and so it's also interesting scott about instead of going from the um non-living systems and then looking for that pattern and pattern matching to biological systems we can kind of start up on the top left with or something even more biologically fleshed out and then look for those patterns and pattern match outside of systems that have you know a vertebra and a central nervous system because that shouldn't be the um the card to get into the club of information processing systems or entropy secreting systems so any thoughts lance yeah i mean yeah i think i think that's that's really interesting um i mean the the the real challenge here is to is to make the link between the physics and the information the the more i've been working on this the more i realized that actually a lot of a lot of physics or at least a lot of statistical physics you can um you can explain with properties of of information so for example entropy increasing is also information being lost not not stochasticity stochasticity so the inability to predict the future exactly and it is exactly the same as entropy being lost over time information being lost over time and entropy entropy increasing so i think there's um there's very deep connections between um yeah information physics uh inference and uh so some of them have been fleshed out but not i mean it hasn't it hasn't really been put together at least as far as i know in this context in the context of markov blankets in the context of how how can we like have a theory of how adaptive systems work can we describe adaptive systems as performing imprints um so so i think this is this is really exciting and and also as you can see from this paper so the the dynamics that we're considering there they're extremely general and for these dynamics for all of them were able to derive some kind of inference so this is to say that this uh inferential description is it applies to many systems and so it means that on the one hand it's very ubiquitous but then on the other hand it's not it's not that useful because maybe like uh the things that didn't look at all like biological systems can still can still satisfy this kind of uh free energy principle or bayesian mechanics so so i think what's really exciting is to be able to um in in the future to to like specialize to those dynamics that look more like biological systems and then really look at these informational properties and i think a big one is how much exploration do we have and exploration is really something exploration and curiosity it's really something that characterizes uh intelligent systems as we know it and yeah so there's some discussions in in the literature of how we would be able to use things like uh information length and concepts for information theory and information geometry to quantify things like memory and itinerancy curiosity and so on so one question from the live chat before you continue the question is what is the explicit definition of information flow and also maybe in that if you want to describe a little bit about this other paper like what is information flow what is information geometry and how does it relate to what we're talking about here right um so so this this is all the idea of um as as an organism you're always encoding some beliefs about your environment so we talked about maximum posteriori and variational inference so the idea is that we're always receiving sensations and then making some inferences about the outside now if you're making some inferences about the outside implicitly or mathematically you're parameterizing a distribution about things in the in the environment and so information geometry uh then comes into play uh information geometry quantifies the extent to which beliefs about the environment are different from each other so it doesn't it doesn't make sense so for example let's say that um i think uh daniel's um yeah uh daniel's uh eyes are blue let's let's just say and then i have some uncertainty around the real color um so that would be like one one kind of uh belief i could hold and then another type of belief would be daniel's eyes are brown and then i would have some other kind of uncertainty now the way um now a question you could ask me is how different are these two beliefs um and if you have a if i had a different belief about the color of his eyes would that believe be more different than the beliefs i initially had um so then information geometry enables you to answer these questions it enables you to say how how two beliefs are different from each other uh by quantifying the distance between them but also even more interestingly as i um let's say i i've never met daniel before let's say i haven't seen his picture before now you give me a picture of daniel um i'm gonna make an inference about the color of his eyes so from being initially completely agnostic i'm going to move my completely agnostic belief to a belief that says oh maybe he brown eyes so this is to say that as we make inferences and sample the world our beliefs evolve and so information geometry enables you to say um how it enables you to quantify how much your beliefs evolved evolved and what kind of trajectory did they take because i i mean i'm assuming at least our beliefs uh they they evolve in a continuous manner as we sample more information and then it becomes interesting to see uh what kind of trajectories did my beliefs take and was that was that trajectory efficient uh if it wasn't efficient maybe um i lost some energy there that i could have used somewhere my my inference wasn't very optimal from that perspective and then so there's a paper about this actually uh on active inference that uses information geometry to to see how how um whether inferences in active inference are efficient from a metabolic point of view so something that you can look at with information geometry thanks just to follow up from the chat and then we'll return to our colleagues here do we have um or more precisely do we hope to have a definition like an electromagnetic flux in field theory for information flows information flux yes um yes so so information flux is the the movement of beliefs and so you can you can write that down mathematically so so yeah i mean a belief is a probability distribution about things that are going on in the and as you sample information you would change probability distribution so it'd be like a motion on the space of probability distributions that's really the the meaning of information flow but maybe the question was about um can we derive some kind of like maxwell's equation some kind of law about how um about how information flow happens like maybe an equation for information flow and i would say that that's what we that that would be the aim uh that would be what what people are working towards in theoretical neuroscience like some kind of equation that explains how we make inference so a few raised hands scott first then blue then stephen so go ahead scott a couple of things that was fascinating little bits there um so on that last piece on the information for those two points um the embodiment i'm thinking of embodiment of information so i'm thinking of a farmer that's deciding what crops to plant so they look and they get information from the newspaper and they look at weather reports and they gather information then they decide on what seeds to plant the seed itself is an embodiment of information they put the seed in the ground they get results with their farming the markets then respond so there was there were seeds as information farmers processing information markets processing information and one of the things that's kind of interesting here it's a lot of the work i'm doing now is in food chains and trying to figure out how to de-risk food chains and and just food generally that efficiency question you just asked is really interesting what are the losses and conversions that happen when information is embodied in those different aspects of that decision process through which something gets converted from a need of society to have a certain amount of corn at a certain time of year to through all those chains of seeds and selections and farmers and processing all those things it's kind of interesting to look at the system as both each piece of it is living the seed is arguably living although there's a question of where the seeds are living among people who are biologists but anyway and then uh because they're not actively doing stuff but the they're kind of stasis but um anyway that embodiment question is that's one and let me ask i'll ask both and then turn off my voice here so the embodiment question is one kind of that notion you just talked about about an overall formula maybe there's an overall embodiment characterization that's made possible using those landscapes and looking at kind of a multi-faceted landscape and looking at those conversions that happen from one embodiment of information to another and efficiencies there that's an interesting kind of possible thought the other one is you said something about curiosity before maximum exploration maximum curiosity that really struck me because one of the things we've been advocating for i do did a lot of cyber work a number of years ago and people were coming out of the military and we were saying oh hey you ex-military guy you got to learn more critical thinking and that was kind of insulting because they were senior people right they're like hey we can work pretty quickly but really they didn't and one of the things i started saying is don't tell people they have to have critical thinking that's not consulting how they have to be more curious and it's kind of interesting that we're getting now to these systems having a maximum exploration possibility a maximum efficiency in some ways with curiosity to me that really informs a lot of teaching because if you have people who develop a curiosity over their lifetime they take care of being educated they continuously are learning and so that notion of exploration and curiosity seems like it could fundamentally through active inference affect how we do teaching and much less of rote teaching and much more of a curiosity based teaching because of the power you'll never know what you're going to encounter out there in the world and so telling your children hey be curious will intrinsically serve them versus teaching them a rote series of things to respond to their environment so just a couple of comments on those last two things i thought were fascinating thank you yeah that that's really interesting um so as you said and i think we all agree i mean information um or maybe 70 70 years ago or 80 years ago we didn't have any kind of information theory information theory wasn't a thing computers weren't the thing and we didn't really think about information even though it was there in books i mean i mean everywhere around us and i think with the advent of computers and information theory just information became a big thing and people realized that it was important now everyone thinks about everything in terms of information and and i think that's the the right way to to look at things i mean it seems to be um yeah information can encode just about anything by by definition um and so physics uh or at least our descriptions of the world they're they um matter was the most important thing matter and and waves uh this is how it all started and then um and then information geometry and information theory came into play and people started looking at these things in terms of uh information as well but i think mata uh remained the big thing but more and more people are looking at at information in terms of like the the most important concept at least the the root concept from which uh from which you can like encode or explain everything else so so yeah information geometry came around and it enables you to to see how how information is flowing over time and how that relates to heat and things like that and now people are looking at how information may be encoded by things about other and this is really basic mechanics based in mechanics the the root of the word is beliefs about things that are held uh that are held by other things and i think this is um yeah i mean basic mechanics i guess started out i mean in a way it started out long ago with like basic imprints but but uh bayesian inference started out as a as an abstract uh statistical concept and now people are trying to relate that to to physics and yeah so so i think we're moving towards the physics that's information based and uh and based about uh yeah based about beliefs about things encoded about other things so this is really basic mechanics information geometry and then what you said about curiosity was also really interesting so so yeah definitely i mean in ai uh people are tuning curiosity so um yeah they're trying to optimize curiosity of agents with respect to different environments in uh in a paper that we have called active entrance on discrete state space is a synthesis we uh in in one of the appendices we we justify the expected free energy objective that people use for uh to explain decision making in biological agents and we actually show that we could have so the expected free energy for for those who don't know is basically ways exploration and exploitation so exploration is really the curiosity so it has these two terms and in general i mean these two terms are put on the same footing but actually in the in the appendix of active inferences synthesis paper we show that you could um weigh them in a different way and and it would and i mean the yeah the agent would still work somehow but it will have a different behavior so you could be a lot more curious or a lot less curious and that would be really interesting to quantify in terms of like uh concepts of information geometry but so so this is to say that that you can actually tune that and have agents that have different curiosities and also make me think about what you said about things from computational psychiatry where um people explain diseases like computational diseases as um as people having like different priors in their genetic model and seeing how that how that would affect behavior so you can explain like sub-optimal optimal and sub-optimal behavior with like the same principles just by changing the model and we also have um a recent paper on the bayesian brains and the rainy diversions where but we're basically showing that by changing doing variational inference a bit differently you can actually um account for for agents that are optimal that make like the right decisions and agents that maybe are a bit sub-optimal in terms of curiosity and and so on so i think there's a like a lot of exciting computational work to be done in terms of characterizing different behaviors and i think we're really at the beginning and ultimately all these will be able to i mean people will be able to account for the information geometry i think that's the right framework thank you lance just quick follow-up scott go for it yep quick follow two things when the beijing mechanics when you say beliefs about things that are held by other things that feels nice with my example of the farmer and the seeds because it's an embodiment each thing that embodies something is the embodiment is a belief in a sense right because it's holding a seed doesn't have a belief itself but it's embodying it's interesting to think about belief of things and then the embodiment of information in the thing intrinsic affordance that the thing offers right that is that what's the relationship between belief and intrinsic physical affordance that's in in a system that is the result of an earlier information system that's one follow-up then the second follow-up is the way i soften people up when you talk to older people and you're like oh it's all information it's not physical we're information beings the way i soften them up just to share is i say to them do you have a 401k or retirement plan most older people are starting to think about that and then they they say yeah and i say do you think it's a pile of groceries waiting for you to retire because it's not it's just data in a system and if that system goes away you got nothing so it's that idea of getting people to understand how much we depend upon information and then people start thinking oh wow i guess i really do depend upon all sorts of other systems in my life it's a something that as physical beings it's hard for us to realize how much we are information beings and that's again just part of that rhetorical exercise to get people in a head space of saying wow maybe things really are information that i thought were solid like my retirement plans but really they're just totally dependent on systems and the information output of those systems thanks nice scott thank you so um blue with a question go for it so you mentioned a paper um that you well i don't know if it was your paper but about information geometry and activated friends is it this one that's markup blankets information geometry and stochastic thermodynamics is that the paper you're talking about uh no that that's not this one so here uh if you're referring to the one um yeah about how um inference is efficient like quantifying efficiency of inference and how much like energy um you need to do a kind of inference um so that's not this paper it's another paper called uh neural dynamics under active inference um something like that so i um have some questions and i don't know anything about information geometry i mean just a little tiny bit um so it's something that i'm definitely starting to get very interested in because i think that um it's gonna be maybe like the next level of where active inference goes maybe maybe i mean i just have this suspicion so when um when you have information and in an information geometry like in an information flow uh i've been reading other papers about information flow in channels right so like when you and i mean it makes sense because when you have when you know something about the current state i mean it's just like an inference like you your belief about the next state of the world is based on the state of the world right now right i mean it doesn't make sense that you would think you know all of a sudden i'm going to be like on pluto based on my current information and so it does is information geometry or do the channels appear when you look at inference in information geometry like does it appear to be like a trajectory that's a good that's a good question so i think i think information geometry is is more general like you don't need to have a channel it's it's really about at the heart of it it's really about having different probability distributions and these are actually beliefs about stuff and looking how they differ from each other and like movement uh how how um probability distributions change over time these kind of problems um but then um so so you're right to say that information theory started out with the study of channels communication channels between different uh different things um and you have a yeah there's a lot of work uh on information theory and channels and still today with uh in engineering i would i would guess there's a lot of work also on information geometry with channels but that would be like um a subset of the whole field of information geometry and i would say it's uh it would be yeah i would say the people contributing to that literature would be mostly from the um from engineering or around engineering thanks blue anything there yeah go for it yeah if i could just follow really quick so the um so i know like about information theory and like the flow of information in the channel say like a telephone line or you know whatever like some direct method of communication but like the way i i don't know i don't know if i'm thinking about it in the wrong way because when i try to search for like information flow in a channel um and maybe it's the same but i think about it in a different way like i think about the channel flowing from like the present into the future right so like there's only so many like next states that are possible right like based on my present state and so is it is it incorrect or is it correct to think about like not a direct channel as in i'm speaking to you over this internet information channel but like the channel from now into some future state is that the same kind of like mathematical engineering information channel right um yeah that um yeah you're catching me off guard because i don't i don't know about channels um so i mean what you say sounds sounds possible and i would say yes but i wouldn't be able to say for sure just because i don't know that literature thanks well i'm sure come back to this in the weeks and years to come so stephen go for it yeah i'm i'm really interested by the um these different um tying these bits together in a way i suppose that sort of come up um we've got this sort of information geometry that i suppose can come from an ergodic space or a kind of a state space and then you've got solenoidal flows within maybe an iso contour of that or potentially i suppose it doesn't have to be solenoid or it could be like an iso contour flow if it was more complex when you've got that potential to um go and have a solenoidal type flow to sample would that be um more a slightly more of a second order sampling approach that something might use so you've got you know the distribution could just emerge from a steady space steady state sort of ergodicity process and you would maybe somehow systematize a flow or way of engaging that information and i could i could imagine that would then make sense of some of the entropy entropy consuming potential of a living system which is that it can kind of go upgrade in descent and the solenoidal flow then gives scott's sort of entropy secreting option it seems a bit more viable then because maybe i could choose multiple solenoidal samplings of information geometries and secrete the ones which aren't as good if that makes sense so um so i'll be curious in terms of how you know that this the ability to go to more second order understandings within a system um and within the dynamics might be offered by the solenoid or sampling maybe even taking into account things like the the landscape used on the um the mountain car problem you know that landscape if you can't sample the whole landscape ergodically could you have lots of solenoidal samples or something like that around the landscape yeah it's a really good point so there's deep connections between so you know flow and information geometry and mostly in the context of the as you pointed out the sampling literature so in um so the the idea is that if you start a stochastic process somewhere and you have solenoidal flow it will converge to its steady state a lot faster than if it didn't have solenoidal flow so then you can quantify that in terms of uh information geometry maybe taking a shorter path to the steady state um and things like that so then there's also um a really nice new paper actually called um is there is there an analog of nesteroff acceleration for mcmc and so that that paper shows that yes that paper really like does a nice job of illustrating some some deep connections between uh stochastic processes and information geometry and so the idea is that if you start with a very simple um called a very simple process called overdumped knowledge of dynamics then you can show that um actually the the dynamic in information space is doing um so like the process is doing a gradient descent on kl divergence um to reach to reach the the steady state so the target distribution so this is just this process is exactly doing like standard standard variational but then if you add some solenoidal flow you actually make that convergence happen faster so it's like an accelerated method and so and so in terms of biology you can say that if you have solenoidal flow um your inference happens faster so you're like uh you're able to infer things a lot faster so you like in a sense more intelligent i think we talk about it in in one of the figures in in the part of the paper and so and so in that paper that i just mentioned uh they showed that with a particular stochastic process actually implement some kind of accelerated variational inference so yeah there's some literature about that and it would be interesting to see uh those processes that perform inference very fast whether when we put them in the bayesian mechanics contest whether we can say oh these actually can be seen to correspond to to biological systems that we've been studying things like that okay i just asked one quick question on when you say add in solenoidal flow is that adding it in the sampling methodology or is it somehow maybe i suppose it's a bit could it be thought of physically or is that how does that work yeah definitely so the the sony nodal flow is really this time irreversible part um so so intuitively it's this kind of like stirring or mixing the system so maybe if you have um a really nice example actually is if you have a cup of coffee and you and you pour milk in it now what you could do is just wait for the milk and the coffee to mix together and this would be no solenoidal flow because you're not mixing anything like it's just happening on its own and you have a dynamic like in the in the figure that we saw before like on the bottom left so it would be time reversible and this would like actually converge the steady state pretty slowly so in terms of information geometry you would have a slow dynamic to the steady state in terms of statistics you would have a slow variation of inference because you reach your target distribution very slowly now what you can do is adding in solenoidal close so this would be i take my spoon and i'm going to stir the coffee and the milk together then we all know then that this is going to allow the system to mix and to and to actually become uniformly mixed a lot faster so so mathematically it means that we reach the target distribution of the steady state faster so the information geometry we're also faster in terms of statistics you can interpret that as doing variational inference faster now this interesting thing about adding in solenoidal flow is that um in physics it's called breaking detail balance and breaking detail balance is about um you actually probably need to to input from entered input energy from the outside um so so this is the physical meaning of the time irreversible part of the dynamic when when there is sometimes irreversible part it means that the there is actually energy coming in from the and and this is also like um it ties in nicely with biological systems we know that biological systems consume energy from the outside so we know that they have to be that they need to have some kind of this irreversible dynamic thanks lance so anyone else in the live chat who wants to ask a question this would be a good time to go for it and then dean and then anyone else let's be gentle on me what i'm about to put out there is i just feel really vulnerable but i'm gonna throw it out there anyway so lots of chat chatter about the encoded aspect of this and i think we all kind of understand what that means i'm gonna i'm gonna raise a metaphor kind of a around negative space and the idea of threading a needle which is if you think about it as a just as a distributive exercise is plenty complex like there's a lot of math in terms of being able to just do that thing that's that's hard hard enough if i'm trying to explain to somebody else a bayesian manifold and what that represents can i say that it's both a barrier and a filter at the same time can i say that it acts kind of like a sieve meaning meaning that it's mathematical structure kind of acts as a ground to the physical things that we are inferring are going on in the world so i guess what i'm asking is is statistical mechanics serving allow us to understand what negative space is as a non-form within all as as part of all the forms that we're seeing the edges of and and having a liminal understanding until we get to the very limits of what and the high definition of what that means does the statistical manifold allow us to go in around and pass through zero it doesn't allow us to reverse the spin of the solenoidal but it does allow us to get away from zero and move closer to what the known if i if we continue to kind of go back and forth between the physical math and the statistical math i know why we have to kind of talk both at once like we have to have two hands to clap but i think it's i think what ends up happening is in this conversation the encoding and the physical piece tends to dominate and then and the non-physical stuff the statistical stuff maybe that ground that the statistical manifold allows for it doesn't it hasn't maybe been fully fleshed out in a way that helps me help explain it to other people who aren't as far along in this journey as you are and i'm hoping to get to a place where i can translate it for them so maybe you can just talk to that a little bit about what is the statistical manifold is it just the hard stuff that we see or should we be paying attention to maybe the spaces between all of those data points yeah so that's a a really interesting and actually a hard question to answer um but just to start with the the easy part the the statistical manifold is the mathematical object that that says these are all the beliefs all the different beliefs that you could hold about your environment right and so as time as time evolves and you're gathering new your what your brain is doing is moving on this statistical manifold your beliefs are changing over time and then in terms of um explaining it to people so the always the the way the way that i try to explain it is that this um this like in information talk um things encoding things encoding information about other it's just a different description of the world so we have a description that that takes like matter and waves and forces and things like that and then we have a different description that looks at and i think these two descriptions are equivalent um it's just two different ways to talk about the same thing on the other hand this um way of talking about the world in terms of information flows and information processing i think it might be more general in the sense that instead of needing to have like matter and all these different concepts different kinds of particles and it gets really messy maybe it would be possible to unify more with the concept of information because with information you can explain what a photon is and what what a proton is and and you name it so so it would be more um more unified and maybe more mathematically simple at the end of the day but that remains to be seen um interestingly so so there's this guy evolved from from uh the computational engine evolved from alpha so this guy this guy's actually a physicist i didn't know that until very recently but he has been doing some really interesting work over the past few years working towards a theory of everything that is based on the concept of information so information and computation is the like the the thing he starts with and i think he's managed to um he basically postulates that the that the um the world evolves in time according to some kind of computation that's recursively applied to itself you you could think of it in terms of a dynamical system but so it's a like the way he thinks about it is a bit more general but it would be like maybe um you give a number to a program and then the program outputs a number and then so that would be one time step and then you reapply the program to that number that would be another time step and so he can he and so he postulates that the universe works in that way um it evolves um according to like discrete steps with a repeated program and the program would be the laws of physics it's like kind of an abstract informational way to look at it so yeah so he he makes this um this very abstract approach to physics and uh and and also very simple i mean very kind of unified approach to physics and i think he's managed to derive approximations to the equations of relativity just with that so with a certain kind of program i mean anyway he's been um doing some really interesting work trying to unify known physics with the concept of information and he's written a very nice uh article like a blog post about it i think if you if you google like a wolfram theory of everything you might you might come up it might come up um but yeah it's um i mean it's not it's not technical i mean it is technical somehow but it's not like there's no equations it's conceptually i think conceptually hard what he does but the way he explains it is nice so yeah i think we're moving to this kind of information-based physics but doesn't mean that the physics that we know with matter and so on should be discarded uh it's just like different different levels to talk about it just like we um psychology that talks about the brain uh from a like very overall perspective and then we have like yeah talking about like maybe biases uh our biases decision making so that would be one description of the world and then we have molecular dynamics that will look at the brain from a very from a basic molecular level and and all these descriptions have to fit together somehow and i think depending depending on what we're trying to ask uh one description would be more fit than the other so if i'm studying a human psychology is probably much more useful than molecular thanks lance one quick comment on the information is it allows us to drop into a mode of matter and organization as um which is still within the objective stance but also it allows us to model ourselves and our beliefs and thinking about them informationally about other beliefs or about matter which is a way of mathematically complementing the relational stance so it also moves us towards wolfram's vision of what new science would look like because we actually have quantitative relational techniques that help us deal with our uncertainty and about the way that we interact with the world actively interfering and measuring and manipulating not just like us walled off as the investigator trying to get to the most exact framing of a system so it's really interesting developments so we have time for some more questions so scott and then after that we'll go to steven so go ahead scott a couple of little thoughts on that last bit so that that multiple dynamics that you were talking about very interesting it got me thinking about that coast of england problem from um fractals where if you measure the coast of england with a one mile yardstick versus a one inch yardstick if it's a one inch yardstick the coast of england is longer right it's because you're measuring more than nooks and crannies right in the fractal uh coast so that scale dependence when you talk about the dependency we see nature and see biology through different lenses it's kind of interesting if we have a unifying lens maybe you start to have a fractal and scale independent descriptors then we'll pop out of that because you'll start to have things that'll anyway it feels like this skewer coming at descriptions of systems as you were saying from different in different ways they're the same system ultimately and so there'll be an interesting analysis on the way in which they come together i guess is that this one piece another mention about information and the um that kind of wolfram idea that expansion um one of the things i had a conversation with a couple of um astrophysicists and i was talking to them about um seth lloyd out of mit a part of his book called programming the universe he talks about the idea that all interactions since the big bang have increased exponentially just the number the volume of interactions i was talking to somebody about it and they said well if the universe is made of information then the exponential increase in information maybe that's what dark energy is we perceive dark energy that expansion force as being something but really what it is is the universe expanding out exponentially because it's made of information and so something just to think about it kind of leaps around there between physics and information theory but there's um that's one other piece of it and then the other piece of it was this idea the outside energy you were talking about before the need for outside energy one of the things that i've been fussing around with is whether in fact and in 2013 i did a presentation mit called entropy accounting and then the year after they did something called entropy engines and the idea that was that was that in information theory in carnot's equation you need a hot and cold differential to perform work um carnot's equation those differentials are identical to arbitrage differentials you need for a market to perform to move if there's no information differentials you don't have trading in markets and so one of the questions is roger penrose wrote a book recently about the time before the big bang what he talks about is we have the yellow light of the sun come in and the infrared light go out we don't overheat except for climate change and carbon dioxide blankets because the net energy gain is zero right energy and energy otherwise we'd overheat he said what we gain is neg entropy because the yellow light of the sun has a higher information carrying capacity than the red light that goes out plants construct themselves from the yellow light of the sun create neg entropy and then we projecting forward are now burning that fossil fuel that stored up that neg entropy and releasing disorder in the form of now climate change disorder political and social disorder so where i'm jumping around there is suggesting that disorder accounting and entropy accounting when you have von neumann and shannon's link of physics and information and other links like we have here that maybe when we're talking about entropy as disorder we can talk about it not by analogy but direct application to social contexts and start to use information and understand disorder in society and in markets as uh amenable to the same kind of math that we find in some of these biological systems thanks just a few thoughts thanks scott lance if you want to add anything sure um yeah that was really interesting and um yeah so um i want to add actually that uh all this um the thing that we talk about about like reality can be described at different scales and i think there's a nice extension to this work that that speaks to that you could um if you look at a human i mean it's composed of a lot of cells and each cell has its own boundary and then these cells form organs that themselves have their own boundary so there's um let's read this recursive aspect of like markov blankets being assembled together at different scales and i think um i think it would be interesting interesting to impact that because i think that's how biological systems are formed but also societies are formed as collections of multiple different individuals that are themselves that themselves form organizations that um that talk to each other and form together a society so it's really this recursive aspect and being able to hopefully use the same kind of math to talk about all these different aspects and also i mean people um it's true like a lot of people in the past maybe 10 20 years maybe more they've used the same kind of dynamical models that we have in this paper i mean so very very standard models to model for example interactions between people and in particular models of opinion formation so for example maybe let's say we're in the us and some people are democrats some of our republicans some cannot really make their mind yet and so they would talk to each other they would interact and then they would converge onto a where some yeah some some people would be die hard democrats in that city state so they would kind of stay in that area some people would be die hard republicans and some people would like yeah fluctuate flock to it in the middle i i think we can use this math about stochastic processes and information is really the yeah the the ground i mean the a good framework or useful framework to to model many kinds of things but then i think what a lot of these models are missing is really this belief based aspect so a lot of these models of opinion formation there they just say oh i'm a particle you're a particle and we interact just like particles in physics with interact um so so it's quite simplistic um so i think once you i mean uh things are very small maybe we can describe them in a very simple way um but as you as like marco blankets self-organize at different scales and as you go to larger and larger scales things become a lot more complex and i would say maybe we are the scale at which things are most complex or molecules tend to be quite simple we tend to be quite complex and then if you go up and up again maybe to planets then things start to become really simple again because all these random fluctuations are kind of average average doubt and then we have newtonian mechanics where everything kind of looks deterministic so i think there's this like sweet spot we're in where we're not too big and not too small and interesting things happen there and that's where i would say that's where i would say bayesian mechanics um is hoping to fit in in terms of uh explaining and describing what's going on cool so stephen go for it that that um that idea again that brings us back to embodiment i suppose because we're embodied at a particular size and scale and um they sort of say if you go right down to the smallest scale of like and then right up to the biggest scale we are kind of in the middle ironically but so i was wondering how sort of following on from what you're saying and bringing in the computational psychiatry piece that you sort of alluded to to this is if we if we say that we've got everything that we perceive even the physicality of the world we actually only have access it through information anyway even if we are actually accessing the the true physical world so we've got this process of sense making that we're doing to to make that idea of what is our consensus reality so if our kind of working space that we're in is our consensus reality what's interesting is that with psychology and social sciences you can almost say psychology is a bit the predictive processing model of the world it's like these are predicted ways that are out there um that have been chosen as the models within a particular sort of discrete spaces and those are externally managed so it's like an externally managed system entry percent there could be an entry repeat secreting system where it looks like there is although entropy is not actually a substance to secrete per se and you could say that it's externally managed information geometry that the social science as well with the averages and the psychometrics that they reference to they're all sort of housed out there in the world and you've kind of got the kind of what we see is our reality in the phenomenology of our experience is kind of the boundary but what your work's doing and what computational psychiatry does and which we're not able to do without those tools is it allows us to go into that neurobiology and the actual like the internal internally managed stochastical dynamics of our own models so to speak so not the ones that we make out there to be our perspective on which we try and reduce free energy around but just what is the models that we're doing what you call entropy consuming i we're actually just swimming in that entropy and finding the variations right and so i'm curious like that that that ability for this type of work to contribute to a new kind of um way of knowing about what it is to be in the world and go into our internal models rather than be reliant on these external psychological and social science models what's your thoughts on how this might link into developing that that field so are you talking specifically about the link between all this physics and and the psychology or or is your question a bit more general um well maybe the the boundary between the kind of like the scales of basically the boundary of going for externally making models of the world and standard models like psychologists are doing and social sciences are doing and psychometrics and the ability to actually go into our own models our own phenomenological neurobiological models and just say okay what are the that are alive not what are and how we how are we actually managing not how do we manage our models of what we think they are which is kind of what happens in our kind of niche but we're able to probe our own phenomenology so i was wondering how you see that tying into that yeah um yeah that's a that's a really good point so so as you as you point out i mean now models in like psychology and and uh computational psychiatry i mean they're driven by intuition they're driven by by experiment and they're driven also by your own biases and our observations about um yeah patients who who suffer some some kind of um yeah some kind of let's say schizophrenia and so we know we know schizophrenic patrons behave like this and like that and so we try to create models that kind of reproduce that and then we try to um to validate these models by saying oh uh maybe i can simulate some kind of uh action potentials of some neurons and then i'm gonna check whether actually these organisms that i'm studying have this um same kind of action potential the same electrophysiological responses so i think that i mean i think that's really good um and and the research is doing a lot of progress this kind of work not not this paper but this line of work would be able to to like go the i mean um i mean it takes a different perspective and would be able ultimately hopefully to describe mathematically um what thinking like a human is uh what what is being human mathematically and of course there won't be like a definition of this is a human but they would hopefully be in the future a mathematical description of how we make decisions and what class of models are we implementing to make sense of the world and how can we characterize these models in terms of and and hopefully um the community will get to that through through physical experiments but also through like just mathematical conceptualization um i think using using mathematics you you cannot get you cannot just get there without any like kind of experiment but you still can there's still quite a few constraints um that we have on that we know that that humans have looks for example breaking detail balance um creation of entropy all these things we we know we know humans satisfy like uh i mean they have a lot of um characteristics and we can incorporate that in the physics and i think this really constrains the model space that then people in psychology can go and look into so it just feels like um when you're digging a tunnel and psychology would like kind of dig in one way and basic mechanics would be digging from the other direction and at some point uh they will meet in the middle hopefully um but i don't i don't think they have met really yet and it will take a while what a fun discussion and it's so cool how this paper it's almost like there was a few stairs we we explored some basic ways to frame how nodes on a bayesian graph insulate other sets of nodes and then how that conditional independence can be finessed for these increasingly developed and elaborated networks that do as we talked about expectations over external states all the way up to variational base and then it's like we at the top or at the middle of that staircase started taking it in a lot of other directions so this was really a fun discussion just in the last few minutes maybe we can any and all of us think like what are we gonna continue working on for some of us that's probably a research project others that's i don't know lunch or something else just how are we going to keep on learning about this start to apply it in our own work or research so i have a really good question liz are you working on um next paper where you're going to partition the external state space or have you started or it's just like clicking on a backburner somewhere uh it's not cooking yet so so you can uh you're more than welcome to to go ahead and and start plowing it's at the farmers markets the ingredients are still out there and dave also raised in an email some really interesting questions about other partitionings basically philosophical threads that might be translated into different possible partitionings within internal states or other ways of partitioning because it was the partitioning of the blanket states from kind of one set of blanketing states into incoming sensory and outgoing action active states that allowed for this perception as inference action planning as inference step and it's not that that's the end game of node partitioning so there's so many ways that these partitions can continue stephen yeah i'm gonna be i'm trying to see how i like that analogy the two tunnels or whichever way the maybe it's a whole load of like um insect warrens or whatever it is trying to meet and um so psychology and bringing together some of these processes so i'm going to try and bring that together maybe more actually not so much from a clinical perspective but maybe more from a community psychology and psychosocial psychosocial methodologies that might be used so i i'm going to be chewing on that as best i can so maybe if i ever stuck it might i might fire you an email if that's okay there's many ways we can continue and also i think all along our trails we dropped a few little notes for some other pieces that could come into play if anyone else wants to raise their jitsie hand we can take a last thought otherwise though just in closing lance this was awesome and conor as well for joining the first live stream this was a uh development we all learned a ton so i hope those watching live or in replay also got something out of this paper you're always welcome back anytime you want to jump on a guest stream or be a participant just always welcome thank you so much daniel and thank you everyone for the discussion i really enjoyed it okay see everyone next time thanks bye great times thanks everyone